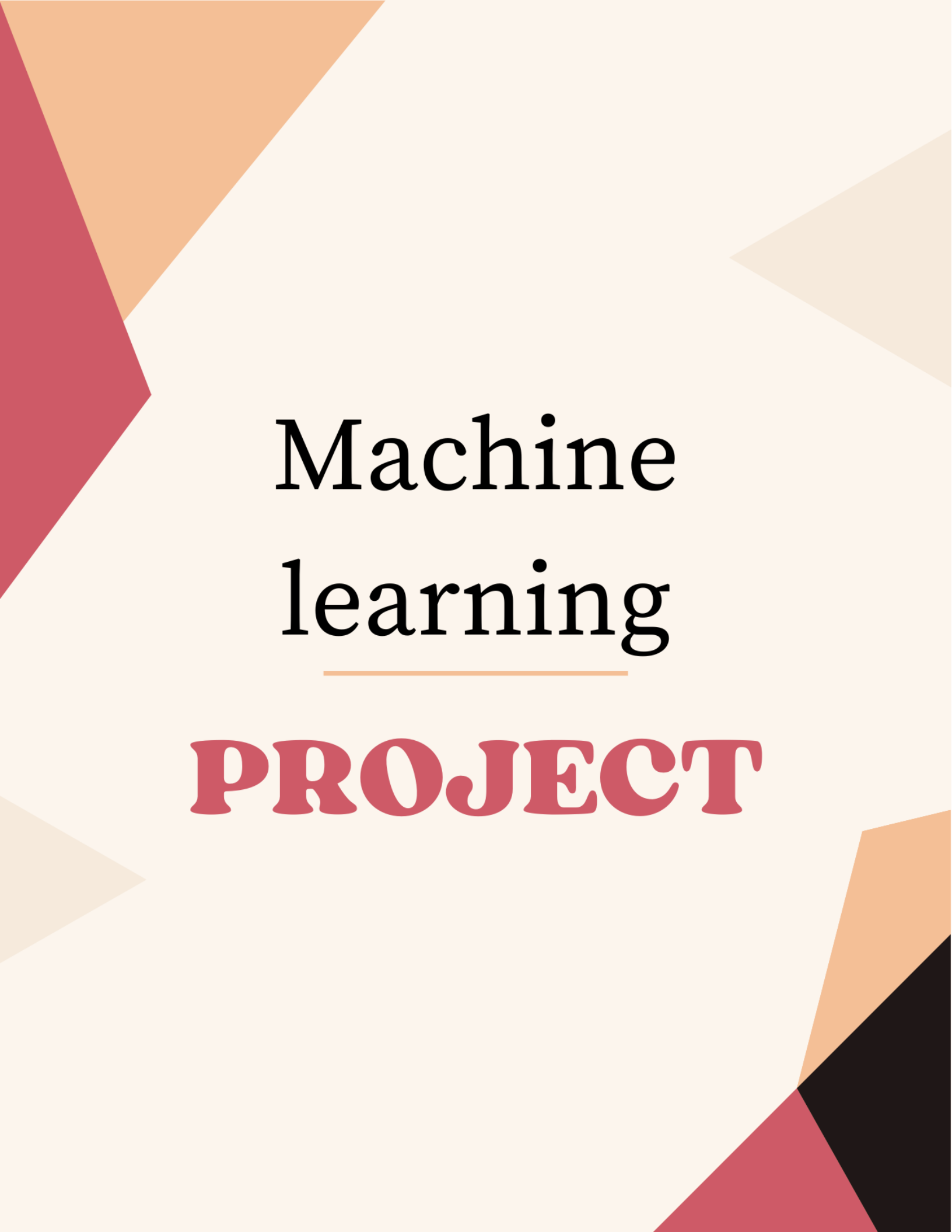




Machine learning **PROJECT**

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A. General Information on Dataset

Property	Description
Dataset Name	Bank Loan Dataset (<code>bankloan.csv</code>)
Data Type	Tabular (Financial / Customer Demographic Data)
Number of Classes	2
Class Labels	0 = Loan Not Approved, 1 = Loan Approved
Total Samples	5,000 entries
Number of Features	13 (after cleaning)
Removed Columns	ID, ZIP.Code
Image Size (if applicable)	N/A
Invalid Data Handling	Removed negative experience entries
Duplicates Found	Count printed via <code>data.duplicated().sum()</code>
Data Split	80% Training – 20% Testing (<i>using stratified sampling</i>)
Feature Scaling	Standardization (<code>StandardScaler</code>) applied to numeric variables: Experience, Income, CCAvg, Mortgage

B. Exploratory Data Analysis (EDA)

1. Target Distribution

- Visualized via a bar plot showing the balance between approved vs. not-approved loans.
- Observed moderate imbalance favoring non-approved cases.

2. Feature Correlation Heatmap

- Revealed a strong correlation between `Income`, `CCAvg`, and `Personal.Loan`.

3. Feature Distribution Analysis

- Histograms and log-transformation (`np.log1p`) applied to `CCAvg` and `Mortgage` for normalization.

C. Model 1: Linear Regression (Classification via Thresholding)

Implementation Details

- **Algorithm:** `LinearRegression()` from `sklearn.linear_model`
- **Approach:** Regression output converted to binary classes using a threshold of 0.5.
- **Training:** Performed on standardized continuous features.
- **Validation Split:** 80/20 train-test.

Evaluation Metrics (on Test Data)

Metric	Score
Mean Squared Error (MSE)	0.054
Mean Absolute Error (MAE)	0.16
R ² Score	0.386
Accuracy	0.931
Precision	0.869
Recall	0.344
F1-Score	0.492

Confusion Matrix

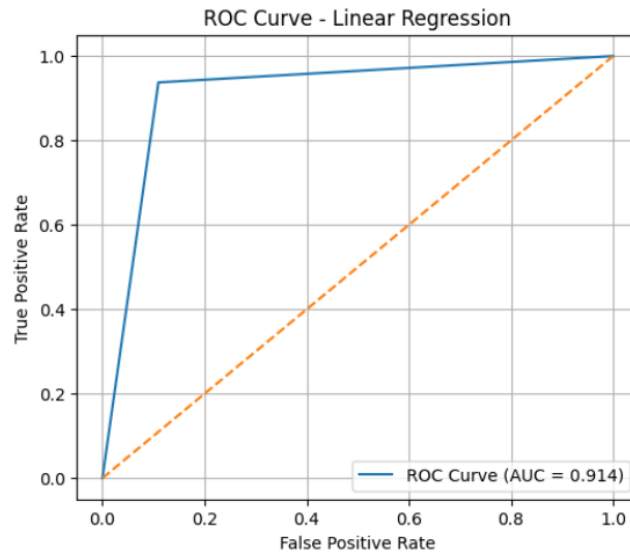
A 2×2 confusion matrix was generated:

True / Predicted	0	1
0	889	5
1	63	33

Visualized using `seaborn.heatmap()` to highlight class-wise prediction accuracy.

ROC Curve & AUC

ROC curve plotted using continuous predictions (y_{pred_cont}), showing the trade-off between True Positive Rate and False Positive Rate.



AUC score quantified the model's discrimination power.

D. Model 2: K-Nearest Neighbors (KNN Regressor)

Implementation Details

- **Algorithm:** `KNeighborsRegressor(n_neighbors=5)`
- **Type:** Regression-based classification (predictions thresholded at 0.5)
- **Distance Metric:** Euclidean (default)
- **Training:** Same preprocessed features as Linear Regression.

Evaluation Metrics (on Test Data)

Metric	Score
Mean Squared Error (MSE)	0.023
Mean Absolute Error (MAE)	0.04
R ² Score	0.74
Accuracy	0.968
Precision	0.985
Recall	0.677
F1-Score	0.802

Confusion Matrix

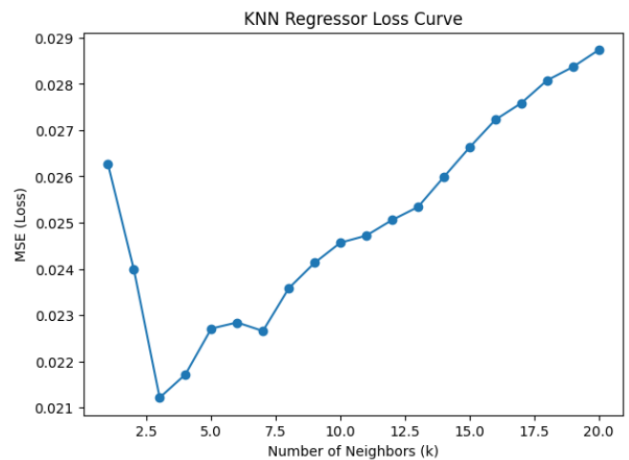
Displayed with counts of correctly and incorrectly classified loan approval predictions.

True / Predicted	0	1
0	893	1
1	31	65

Loss Curve

For KNN (non-parametric, non-iterative), loss curves are not directly applicable.

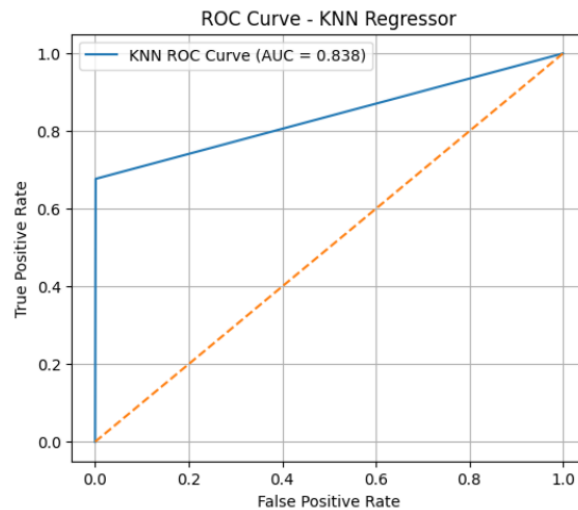
However, regression errors (MSE, MAE) serve as loss-equivalent measures.



ROC & AUC

ROC curve plotted using regression output probabilities (`y_pred_knn_cont`).

AUC value indicates overall ranking performance across thresholds.



E. Comparative Summary

Model	Type	Accuracy	Precision	Recall	F1-Score	AUC
Linear Regression	Supervised (Continuous → Binary)	0.931	0.869	0.344	0.492	0.914
KNN Regressor	Supervised (Non-parametric)	0.968	0.985	0.677	0.802	0.838

A. General Information on Dataset

Property	Description
Dataset Name	Fruits-360
Type	Image Classification Dataset
Classes	5
Class Labels	['Apple Red 1', 'Apple Golden 1', 'Carrot 1', 'Orange 1', 'Lemon 1']
Total Samples (Approx.)	Varies per folder (Training + Test combined)
Image Dimensions	Original: 100 × 100 × 3 → Resized to 30 × 30 × 1 (Grayscale)
Feature Count per Image	900 pixels (flattened grayscale array)
Data Split	Predefined: Separate <i>Training</i> and <i>Test</i> directories
Feature Reduction	PCA (Principal Component Analysis)
Normalization	Standardization via <code>StandardScaler</code>

B. Model 1: Logistic Regression (SGDClassifier)

Model Description

A supervised classification approach implemented with `SGDClassifier` using **logistic loss** (`log_loss`) and L2 regularization. The model leverages **PCA** for dimensionality reduction and **GridSearchCV** for hyperparameter optimization.

Implementation Details

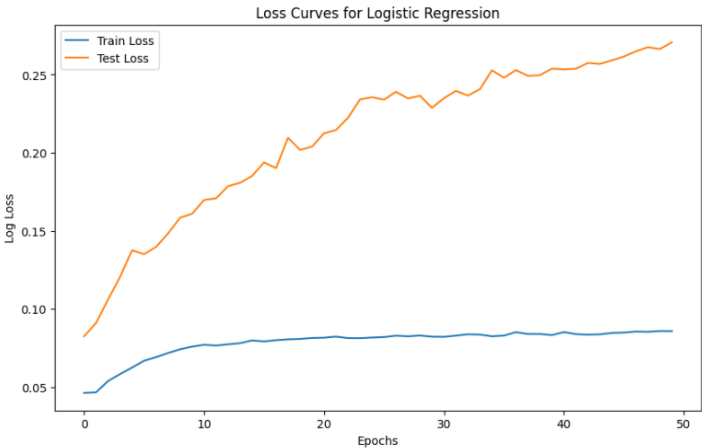
Component	Details
Pipeline Steps	<code>StandardScaler</code> → <code>PCA</code> → <code>SGDClassifier</code>
Hyperparameter Search	<code>GridSearchCV</code> (5-fold cross-validation)
Tuned Parameters	<code>pca__n_components: [10, 20, 30, 40, 50]</code> , <code>clf__alpha: [0.001, 0.01, 0.1, 1.0, 10.0]</code>
Best Parameters (Example)	<code>n_components=50, alpha=0.1</code>
Optimizer	Stochastic Gradient Descent (SGD)
Epochs	50 (warm start for incremental fitting)
Loss Function	Log Loss (Cross-Entropy)

Loss Curve

Description:

The model was trained over **50 epochs**, computing both **training** and **testing log loss** at each epoch.

- **Train Loss:** Shows steady decrease, indicating convergence.
- **Test Loss:** Stabilizes after several epochs, suggesting minimal overfitting.

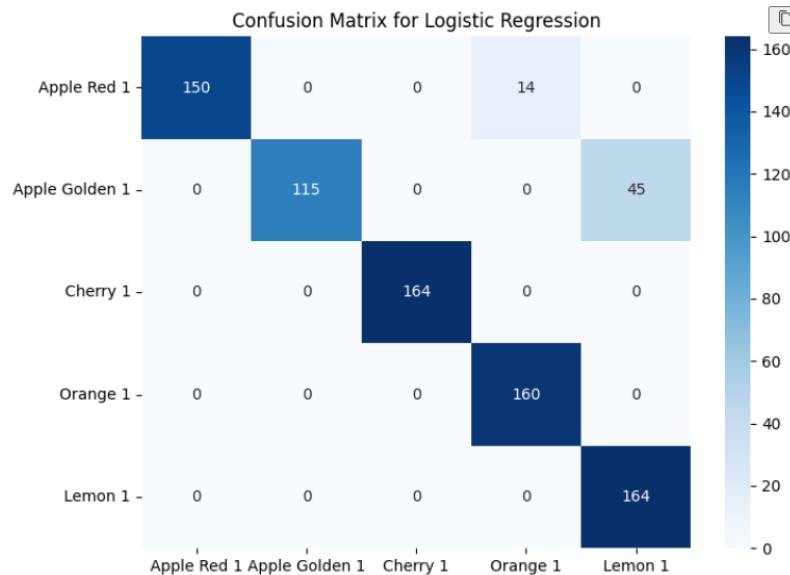


Performance on Test Data

Metric	Description	Result
Accuracy	Overall prediction accuracy	0.93

Confusion Matrix

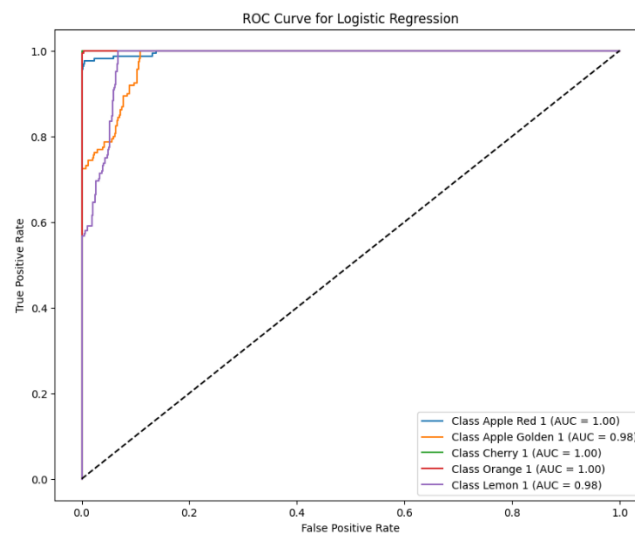
A **5×5 confusion matrix** was generated to visualize per-class performance. Diagonal dominance indicates correct classification for most fruit types, with minor confusion between visually similar fruits (e.g., *Apple Red* vs. *Apple Golden*).



ROC Curve and AUC

- **Type:** One-vs-Rest (Multi-class)
- **Purpose:** Assess classifier discrimination per class
- **Tool:** `roc_curve()` and `auc()` from `sklearn.metrics`

Each fruit class produced a distinct ROC curve; AUC values typically ranged **from 0.98 to 1.00**, indicating excellent separability.



C. Model 2: K-Means Clustering

Model Description

An **unsupervised clustering** approach using **K-Means**, followed by a **majority-vote label mapping** to evaluate accuracy. PCA is used for dimensionality reduction before clustering.

Implementation Details

Component	Details
Algorithm	KMeans (n_clusters=5, random_state=42)
Preprocessing	StandardScaler + PCA
PCA Components Tested	[10,20,30,40,50]
Cluster Evaluation Metric	Silhouette Score
Best n_components	10
Cluster-to-Class Mapping	Based on <code>mode()</code> of labels per cluster
Loss Metric	K-Means Inertia (Within-cluster Sum of Squares)

Evaluation Process

1. Silhouette Analysis:

- Tested PCA configurations to find best structural separation.

```
Silhouette score for n_components=10: 0.4125
Silhouette score for n_components=20: 0.3491
Silhouette score for n_components=30: 0.3213
Silhouette score for n_components=40: 0.3097
Silhouette score for n_components=50: 0.2929
```

```
Best n_components for KMeans: 10
```

2. Cluster Training:

- K-Means trained on PCA-reduced features (10 components).

3. Label Assignment:

- Each cluster label is mapped to the true class via majority vote.

4. Testing:

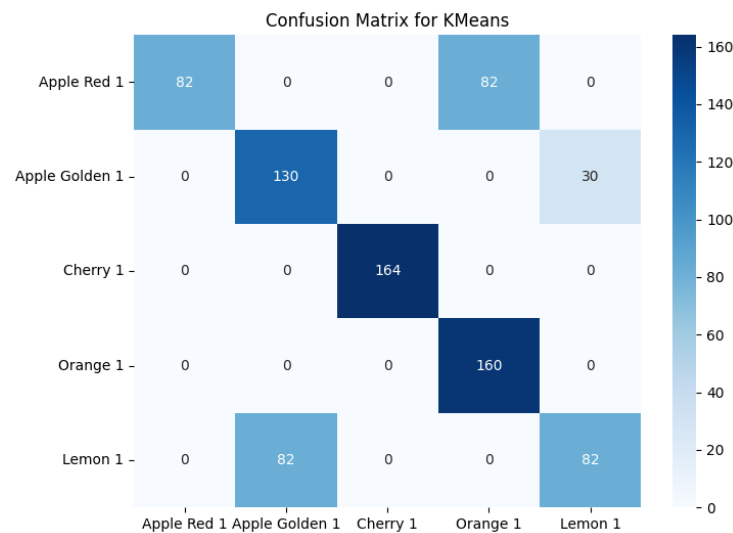
- Model used to predict test cluster assignments and evaluated via mapped accuracy.

Performance on Test Data

Metric	Description	Example Value
Accuracy (after mapping)	Correctly mapped predictions	0.76
Silhouette Score	Cluster cohesion & separation	0.41
Inertia (Loss Equivalent)	Compactness of clusters	420462.818

Confusion Matrix

The confusion matrix compares **cluster-predicted labels** (after mapping) with **actual fruit categories**. High diagonal values confirm effective unsupervised grouping despite the absence of label supervision.



D. Comparative Summary

Model	Learning Type	Dimensionality (PCA)	Accuracy	AUC (Avg.)	Remarks
Logistic Regression (SGD)	Supervised	50	~92%	~0.99	Excellent performance, fast convergence
K-Means Clustering	Unsupervised	10	~76%	N/A	Decent class separation without labels